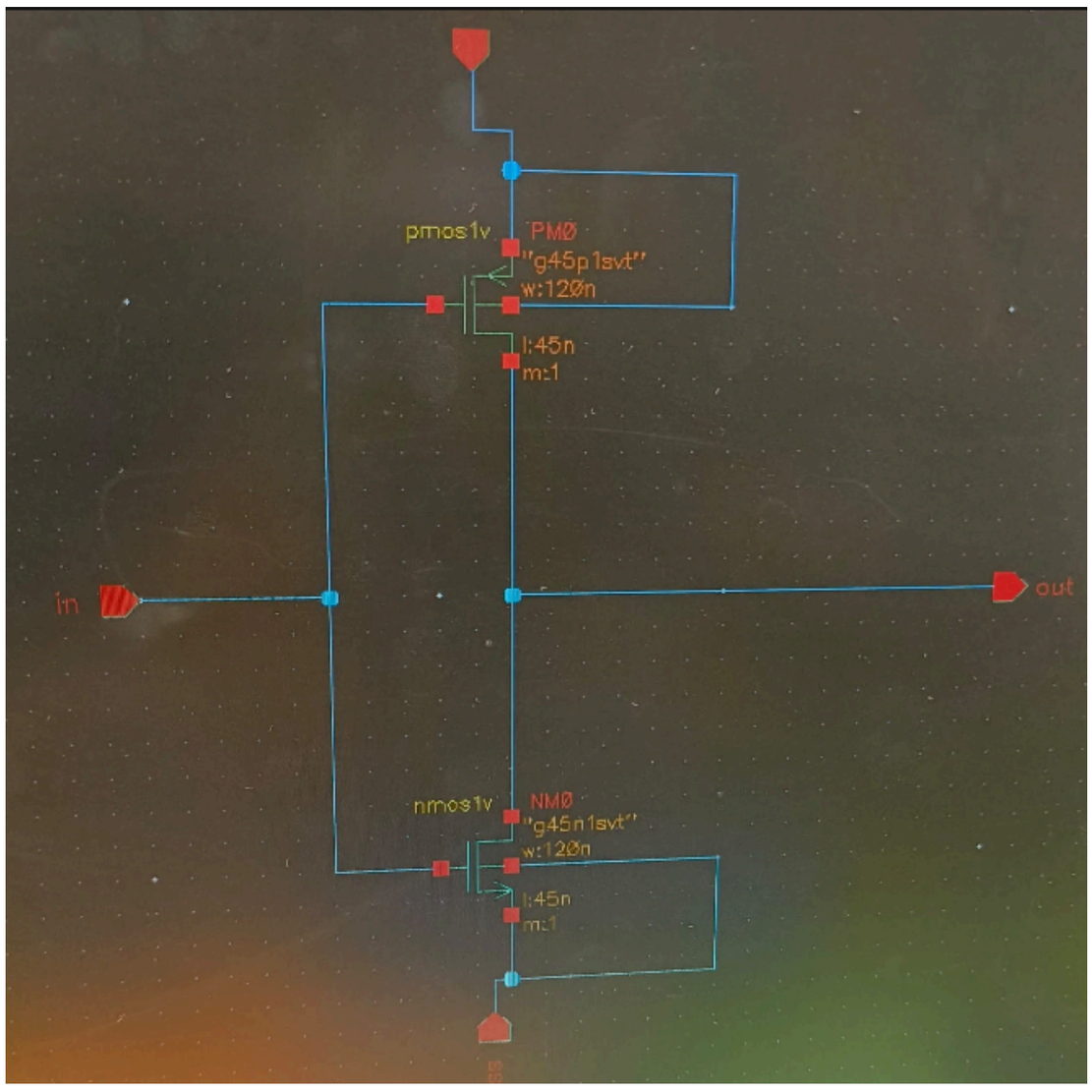
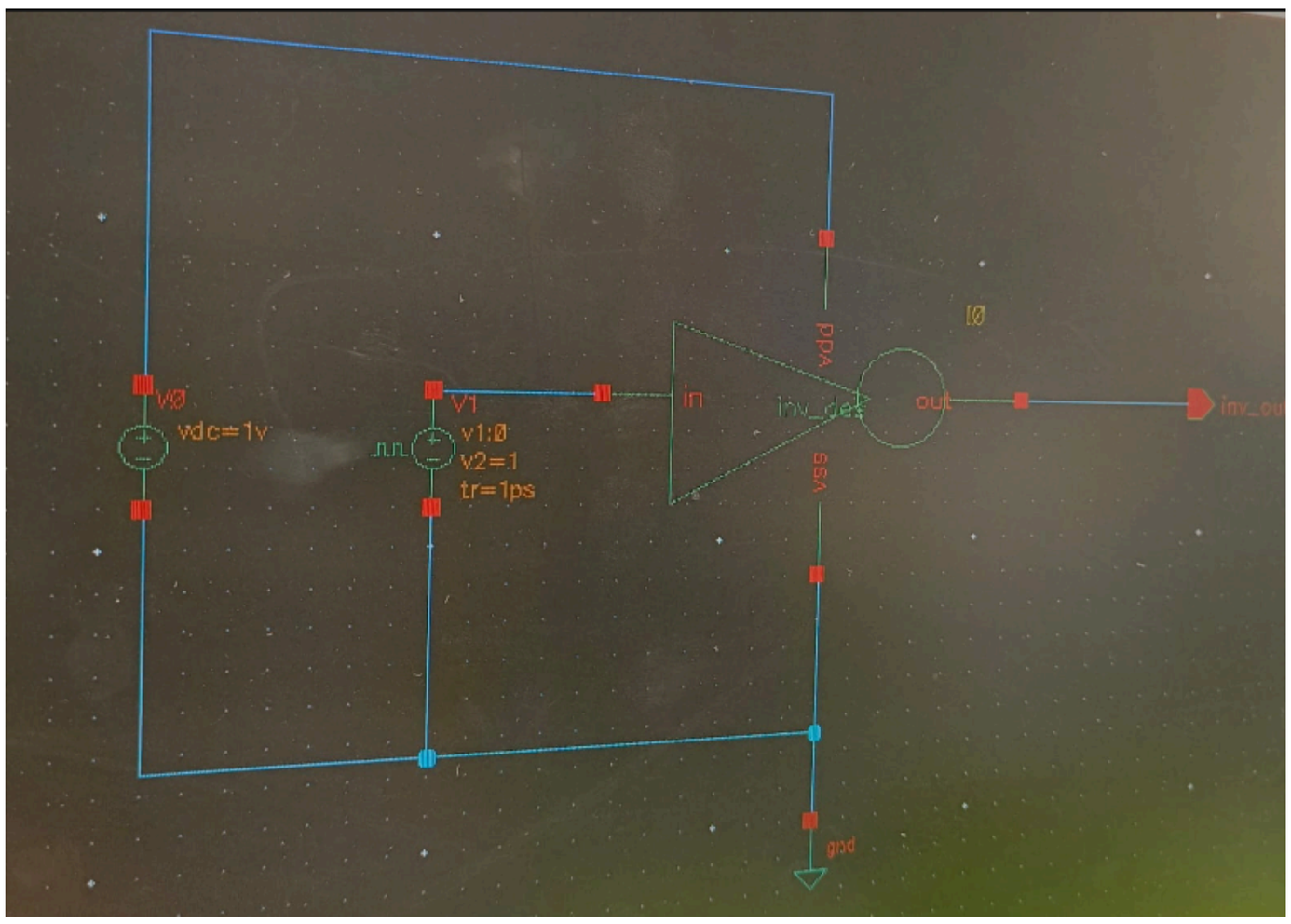
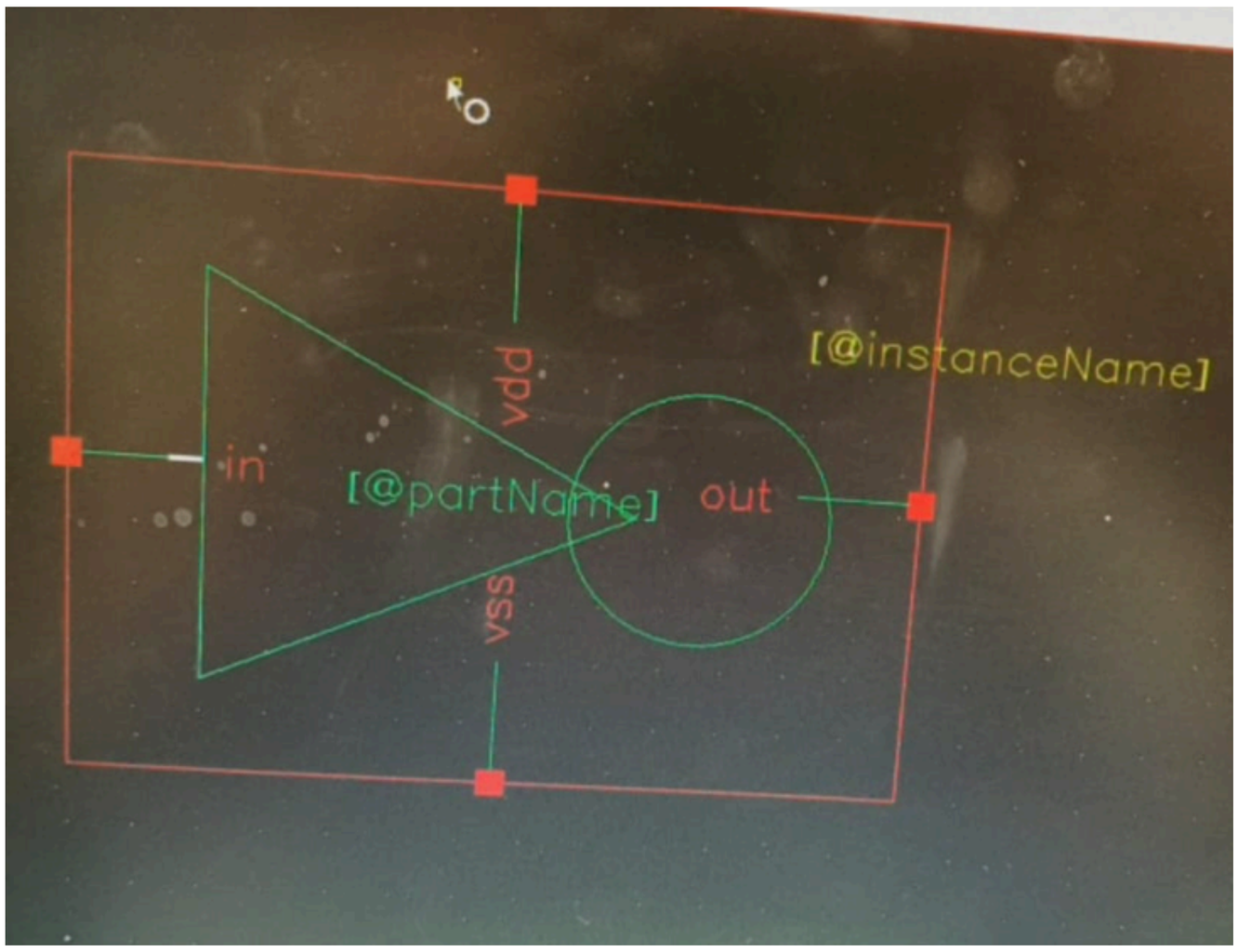
EXPERIMENT 1

Simulation-Based Analysis of N-Well CMOS Fabrication Techniques Using Cadence Design Environment

Circuit diagram:







Output Parameter Table:

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	Fabrication sequence	Standard CMOS process flow	Standard CMOS fabrication sequence observed	0% deviation
2	Gate length	180 nm	180 nm	0% deviation
3	Gate oxide thickness	4 nm	4 nm	0% deviation
4	Source/Drain formation	Properly defined regions	Source and drain regions properly formed	0% deviation
5	Layer structure	Correct CMOS layers	Substrate, gate oxide, gate, source/drain and metal interconnect layers identified	0% deviation

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
6	Device dimensions	As per design	Record simulated device dimensions from Cadence	0% deviation

Output:

